Information Gain Propagation: A New Way to Graph Active Learning with Soft Labels

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Problem

GNNs need many labeled nodes, yet every prior graph **active-learning** method assumes an oracle can name the **exact class** of each selected node.

> *Exact multi-class labeling is costly, especially when categories are many or out-of-domain (e.g. ogbn-papers100M has 172 classes) — the budget is spent inefficiently.*

Motivation

- **Confirming a guess is cheap.** A binary yes/no query costs far less per node than naming the exact class among many.
- **Labels propagate.** One labeled node sways its k-hop neighbors, yet prior criteria score only single-node uncertainty.

PRIOR AL QUERY

"Which exact category?" — a hard multi-class question, ~c−1× the cost.

vs.

OUR RELAXED QUERY

"Is the predicted label correct?" — one binary yes/no → a soft label.

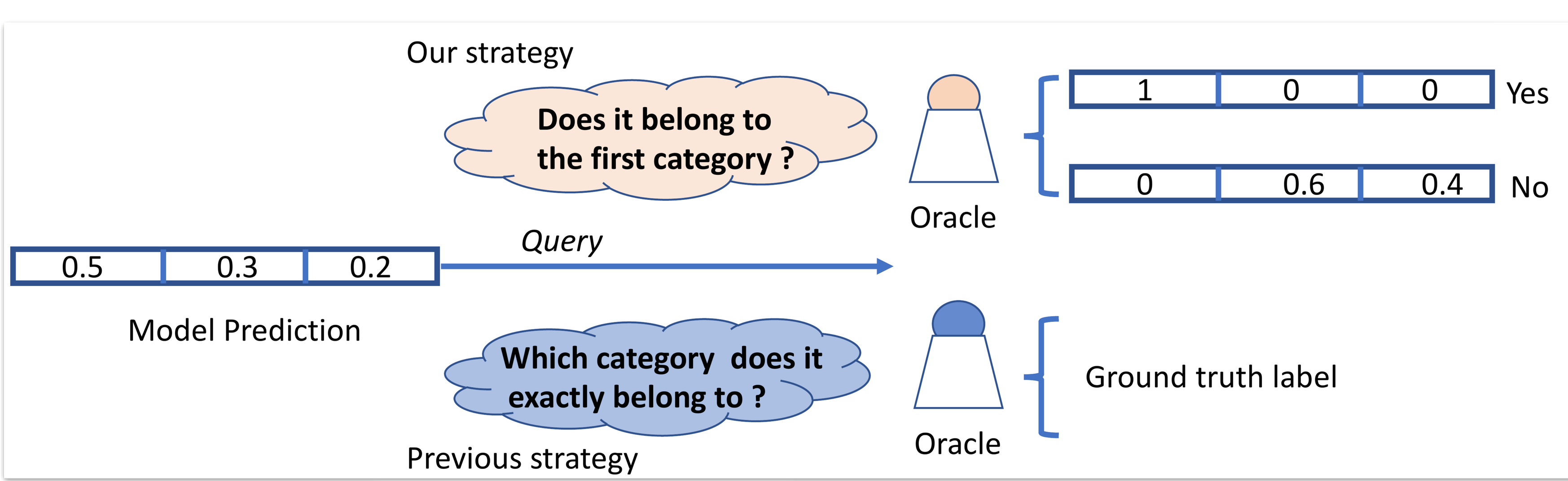

Figure 1: A relaxed binary query yields a soft label, versus the exact-class query of prior work.

Cheaper queries plus propagation-aware selection extract more supervision per labeling dollar.

Method

1. **Train** a GNN on the currently labeled nodes with their **soft labels**.
2. **Score** each node's **influence magnitude** on its k-hop neighbors via graph propagation.
3. **Select** the budget-limited set maximizing **information gain propagation** (IGP).
4. **Relabel** via a binary query → normalized soft label, then update & repeat.

$$\max_{V_l} F(V_l) = \sum_{v_i \in V_l} \sum_{v_j \in RF(v_i)} IGP(v_j, v_i, k), \quad |V_l| = B$$

Select nodes whose expected information gain propagates farthest across each receptive field RF, within labeling budget B.

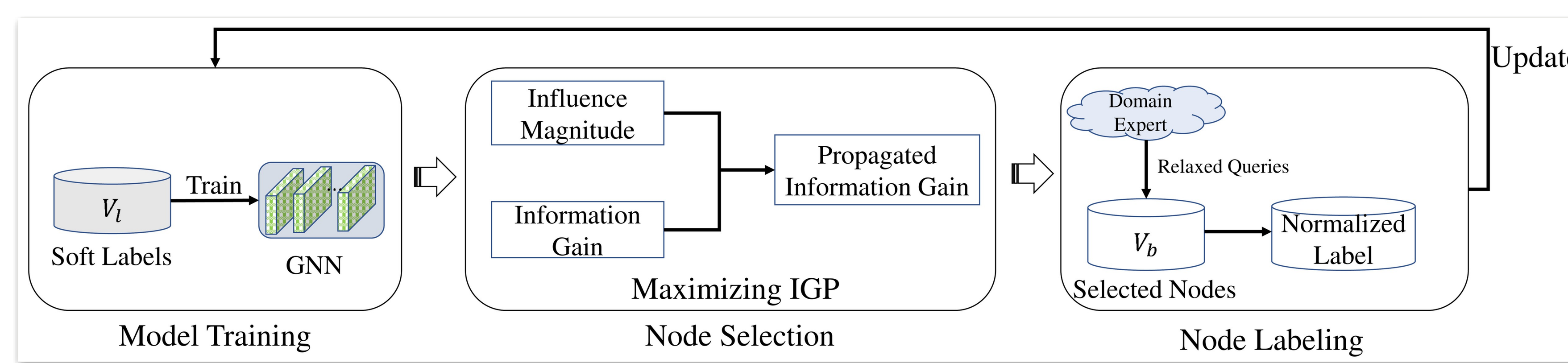

Figure 3: The IGP active-learning loop.

Dataset / Benchmark

- **Cost-based budget.** The budget is true annotation *cost*, not label count — an exact query costs ~c−1× a relaxed one.
- Budgets swept from **2C to 20C** labels (C = #classes) across five standard graph benchmarks.

Cora **Citeseer** **PubMed** **Reddit** **ogbn-arxiv**

Key Results

METHOD	CORA	CITESEER	PUBMED
Random	78.8	70.8	78.9
GRAIN	84.2	74.2	81.8
IGP (ours)	**86.4**	**75.8**	**83.6**

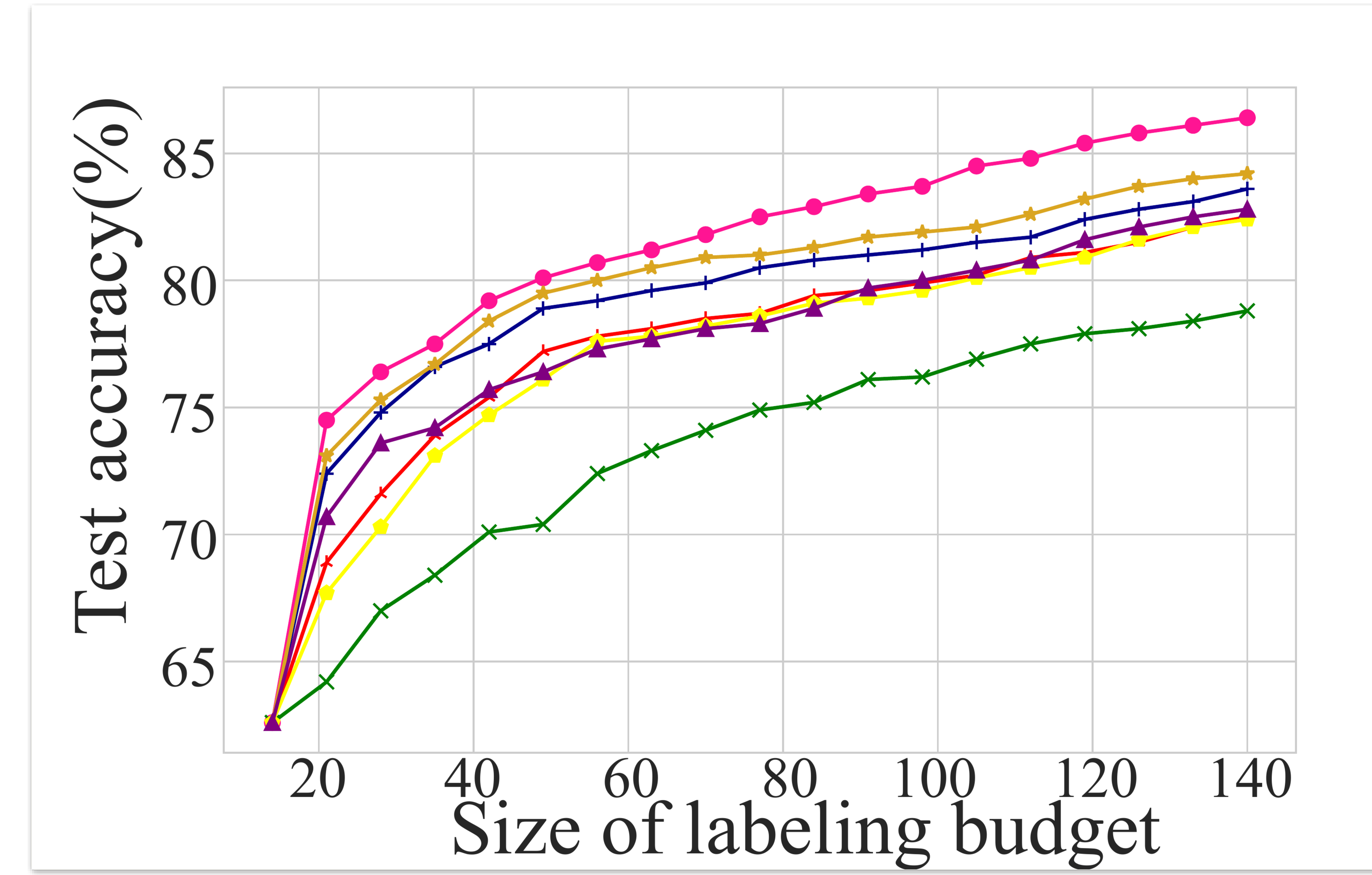

Figure 4: Accuracy vs labeling budget on Cora — IGP (pink) leads at every budget.

Takeaway

Ask a cheap yes/no question and pick nodes whose information gain propagates farthest — graph active learning gets more accurate *and* far cheaper to label, on any GNN backbone.

86.4%	**+2.2**	**5**	**4**
Cora acc	pts vs GRAIN	datasets SOTA	GNN backbones